Report for Problem Set 2

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1. What model we used (SV model)

We worked with a stochastic volatility model. The main idea: volatility is not constant. Some periods are calm, others are very volatile. We simulate returns. But behind the scenes there is a hidden variable called log-volatility. We never observe it directly, we just see the returns. How it works in words: Log-volatility follows a slow, persistent process. If volatility was high yesterday, it tends to stay high today. The persistence parameter phi is 0.95, which means volatility changes slowly and does not jump around randomly every day. Returns are generated using that hidden volatility. When log-volatility is high, returns become more spread out (bigger ups and downs). When it is low, returns are smaller. We simulated 2500 days with settings (phi = 0.95, sigma_eta = 0.2, sigma = 1.0).

2. Why we need the Harvey transform

The Kalman filter only works when the model is linear. Our original SV model is nonlinear because volatility enters through an exponential term. The Harvey transform is a trick to make the problem linear: We square the returns (this removes the plus/minus sign). We take the log of that squared return. After this, the transformed series y becomes roughly a linear function of hidden log-volatility plus noise.

After we make our model liner, we can use it for Kalman filter.

3. What the Kalman filter does

The Kalman filter estimates hidden log-volatility day by day. Each day it does two things: Prediction (before seeing today's data): It uses yesterday's estimate to guess today's log-volatility.

Update (after seeing today's data): It compares the actual transformed return with what the model expected. If there is a surprise, it updates the volatility estimate. The size of the update depends on how trustworthy the new information is relative to the previous estimate.

At the end, we get a filtered path of log-volatility. In the problem set outcome, this path follows the true hidden volatility quite well. That tells us the filter logic is working.

4. MLE results

In practice, we do not know the true parameters. We estimate them by maximum likelihood: Try candidate values for phi, sigma_eta, and sigma. Run the Kalman filter and compute how well the model fits the data. Pick the values with the best fit.

My results:

```
True:      phi=0.950,  sigma_eta=0.200,  sigma=1.000
Estimated: phi=0.952,  sigma_eta=0.164,  sigma=1.107
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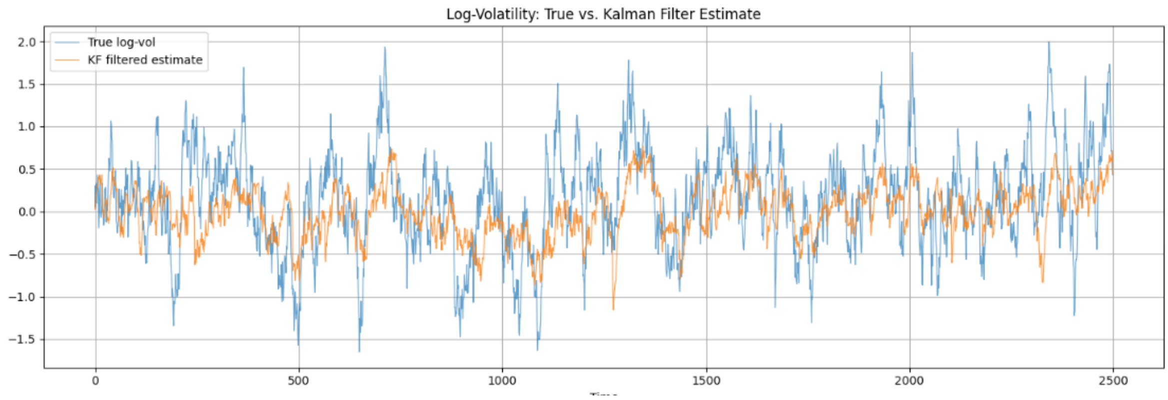

phi is almost exactly correct → the model correctly learns that volatility is very persistent.

sigma_eta is a bit lower than true → the optimizer thinks volatility shocks are smaller.

sigma is a bit higher than true → return scale is slightly overestimated.

Overall, the estimates are close. Small differences are normal with finite data and approximation steps. The important part is that estimated parameters still produce a good filtered volatility path.

5. EGARCH model and what it estimates

EGARCH is the second approach. Instead of a hidden state plus Kalman filter, we fit a volatility model directly to returns using the arch package.

Main logic: Today's volatility depends on past returns and past volatility. Big shocks can increase future volatility. The model produces a daily conditional volatility series

Key EGARCH estimates from my run:

omega = 0.0246, baseline level in the log-volatility equation.

alpha = 0.2357, sensitivity to recent shocks.

beta = 0.9052, persistence of volatility. The high beta (0.905) means volatility clusters over time, which is similar in spirit to high phi in the SV model. So both methods capture the same financial idea, volatility is persistent.

6. VaR and ES

After estimating volatility, we compute two risk measures: VaR (5%): a threshold for “bad but still normal” losses. ES (5%): the average loss in the worst 5% cases (tail risk). Both are negative because they represent losses.

We computed them for both models:

KF path: volatility from Kalman-filtered log-volatility.

EGARCH path: volatility from garch_vol.

On high-volatility days, VaR and ES become more negative (risk is higher). On low-volatility days, they are closer to zero (risk is lower). ES is always more extreme than VaR, because it describes the average of the worst outcomes(the possible lost if we land in that 5% part).

Example from my output: early EGARCH VaR values were around -1.85 to -2.04, which matches the idea that risk changes over time.

7. Coverage ratio

Coverage ratio checks whether VaR is well calibrated.

Logic: Count how often actual returns are worse than VaR. Divide by total number of days. If VaR is truly a 5% risk measure, this should happen about 5% of the time.

My results:

KF coverage: 5.00%

EGARCH coverage: 4.96%

Target: 5% Both are almost exactly on target. That means both models produce realistic risk thresholds for this dataset. Interpretation: KF and EGARCH are both well calibrated here. I would treat them as equally good for calibration in this exercise. In exact number KF is fitting perfectly the 5% expectation but in practice %0,4 in coverage could be ignored therefore EGARCH is also great calibrated here.

8. Conclusion

In this problem set, we compared two different methods for measuring risk.

The first method was the Stochastic Volatility (SV) model with the Kalman Filter. It assumes that volatility is a hidden process and uses the Harvey transformation and Kalman Filter to estimate it. This approach is more model-based and gives a clear interpretation of how volatility evolves over time.

The second method was the EGARCH model. Instead of estimating a hidden state, it models volatility directly from past returns. It is easier to implement and requires fewer steps.

Overall, both methods performed well on the simulated data. The estimated parameters were close to the true values, and both models were able to track changes in volatility over time. The VaR and ES measures behaved as expected, becoming larger when volatility increased and smaller when volatility decreased.

The coverage ratios were also very close to the target value of 5%, which shows that both models produced reasonable risk estimates.

Based on these results, neither method was clearly better than the other. Both provided similar risk measurements and were well calibrated. The choice between them would depend on the application, the available data, and whether simplicity or model interpretation is more important.